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SCROLL MACHINE AND REFRIGERATION SYSTEM

Abstract

A scroll machine for a medium has a machine housing having a longitudinal axis and an inlet and an outlet for the medium. Provided in the machine housing along the longitudinal axis is: a drive unit having a drive shaft mounted on the machine housing by a first bearing unit and a second bearing unit, a first spiral unit having a spiral channel formed by a first spiral rib and has an inner end region and an outer end region, and a second spiral unit having a spiral channel formed by a second spiral rib and has an inner end region and an outer end region. The first spiral unit and the second spiral unit engage one another to form pressure chambers, wherein the first spiral unit can be moved by means of the drive shaft through the drive unit along an orbital path relative to the second spiral unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/233,201 filed Aug. 11, 2023, which claims priority to German Patent Application No. 10 2022 120679.3 filed Aug. 16, 2022, the entire contents of each which are incorporated herein by reference.

[0002] The present invention relates to a scroll machine, in particular a spiral compressor, for a medium, in particular a refrigerant, and to a refrigeration system comprising such a scroll machine.

[0003] Scroll machines are fluid energy machines and are already known from the prior art in different designs. Scroll machines include, for example, scroll compressors, spiral compressors, and scroll expanders.

[0004] Known scroll machines typically have two interacting spiral units, each having at least one spiral rib forming at least one spiral channel. The spiral ribs of the spiral units engage or mesh with one another to form pressure chambers, the spiral ribs sealingly abutting a spiral channel base of the other spiral unit in question.

[0005] Both the spiral channel and the spiral rib forming the spiral channel are in the form of an involute of a circle, with the two spiral units being movable relative to one another. A common design of scroll machine includes a stationary spiral unit and a movable spiral unit, with the movable first spiral unit being moved along an orbital path relative to a second spiral unit.

[0006] According to the displacement principle, a medium, for example a refrigerant, is compressed or compacted in a compressor by a relative movement of the two spiral units. During this relative movement, the medium in the pressure chambers is displaced along the spiral channels from an outer end region to an inner end region, with the medium in the pressure chamber undergoing a change in volume.

[0007] In an expander, the medium, in particular refrigerant, is expanded by a relative movement of the two spiral units. During this relative movement, the medium in the pressure chambers is displaced along the spiral channels from an inner end region to an outer end region, with the medium in the pressure chamber undergoing an increase in volume.

[0008] WO 2018 019 372 A1, for example, discloses a generic scroll machine of the type in question, which can be used in a refrigeration system with a refrigerant circuit. Such refrigeration systems can have a variety of uses, for example cooling a secondary fluid such as air, or cooling components or equipment. The cooling or heating load of refrigeration systems can vary greatly according to ambient conditions, occupancy levels, and other load requirements.

[0009] The scroll machines and refrigeration systems described above have proven successful in the past, but it has been shown that the components of the scroll machine require both continuous cooling and lubrication in order to achieve the required service life.

[0010] This is where the present invention comes in.

[0011] It is the object of the present invention to propose a scroll machine and a refrigeration system of the type described at the outset which eliminate the disadvantages known from the prior art in an expedient manner. A scroll machine and a refrigeration system which are cooled

effectively in a simple manner and likewise lubricated are to be provided.

[0012] These objects are achieved by a scroll machine having the features of claims **1**, **6**, and **19** and by a refrigeration system having the features of claim **23**.

[0013] Further advantageous embodiments of the invention are specified in the dependent claims.

[0014] The scroll machine according to the invention having the features of claim **1** for a medium, in particular a refrigerant, has a machine housing having a longitudinal axis, as well as an inlet and an outlet for the medium. Furthermore, in the scroll machine according to the invention, a drive unit, a drive shaft, a first spiral unit, and a second spiral unit are arranged along the longitudinal axis. The drive shaft is supported on the machine housing by a first bearing unit and a second bearing unit, with the axis of rotation of the drive shaft preferably defining the position and orientation of the longitudinal axis. The first bearing unit and the second bearing unit are preferably arranged in opposite end portions of the drive shaft.

[0015] The second bearing unit is arranged in the end portion adjacent to the first spiral unit, and the first bearing unit is arranged in the opposite end portion.

[0016] The first spiral unit has a first spiral channel which is formed by a first spiral rib and extends from an inner end region to an outer end region. The second spiral unit has a second spiral channel which is formed by a second spiral rib and extends from an inner end region to an outer end region.

[0017] The inlet of the scroll machine is in fluid communication with the outer end regions of the first spiral unit and the second spiral unit; the outer end regions are referred to as the suction region in the context of this invention. The outlet is in fluid communication with the inner end regions, as a result of which the medium is guided through the inlet and through the machine housing along a plurality of flow paths to the outer end regions and from the inner end regions to the outlet.

[0018] The first spiral unit can be moved by means of the drive shaft through the drive unit along an orbital path relative to the second spiral unit, and the first spiral unit and the second spiral unit engage one another to form pressure chambers, the spiral ribs sealingly abutting a spiral channel base of the other spiral unit in question.

[0019] According to a first embodiment of the present invention, the drive shaft has a hollow shaft portion, one of the flow paths being guided through the hollow shaft portion. The hollow shaft portion is preferably arranged coaxially with the axis of rotation of the drive shaft and can partially or completely penetrate the drive shaft.

[0020] The present invention is based on the concept of using the drive shaft simultaneously as a line for the medium, which allows the medium to be guided in a space-saving manner and with low pressure loss within the machine housing and allows the drive shaft and in particular the scroll machine components connected to the drive shaft, such as the bearings of the drive shaft, to be cooled. The proposed solution can provide a particularly compact scroll machine with integrated cooling and efficient lubrication of the bearings.

[0021] It is noted at this point that the medium is preferably a refrigerant, the refrigerant comprising a lubricant which can be entrained by the refrigerant.

[0022] In a development of the present invention, the hollow shaft portion is formed by a blind hole. The blind hole preferably extends from an end face of the drive shaft in the first end portion toward the second end portion.

[0023] According to a further development, the drive shaft can have at least one radial bore, which opens into the hollow shaft portion and connects the hollow shaft portion to an outer lateral surface of the drive shaft. As a result, the medium can be guided along a flow path through the at least one radial bore. The at least one radial bore allows the medium to be conducted out of the drive shaft to individual components in a targeted manner along the longitudinal axis, as a result of which these components can be cooled and/or lubricated by the lubricating medium entrained with the medium.

[0024] According to a development, a rotor of the drive unit can be arranged between the first bearing unit and the second bearing unit, the drive shaft between the first bearing unit and the rotor

and/or the drive shaft between the second bearing unit and the rotor having the at least one radial bore. At least one radial bore is preferably arranged both between the first bearing unit and the rotor and between the second bearing unit and the rotor. The at least one radial bore between the first bearing unit and the rotor and/or between the second bearing unit and the rotor can contribute to improved cooling of the rotor of the drive unit, the drive shaft preferably having a plurality of radial bores around the circumference, which according to an even more preferred development are arranged circumferentially symmetrically.

[0025] In addition, according to a development of the present invention, it has proven to be advantageous for the drive unit or a preferably electric drive around the drive unit to have at least one axial opening and/or at least one axial groove, and for the at least one axial opening and/or the at least one axial groove to define a flow path for the medium. The drive unit preferably has a plurality of axial openings and/or a plurality of axial grooves around the circumference, which are preferably arranged circumferentially symmetrically. The at least one axial opening and/or the at least one axial groove connects opposite end faces of the drive unit or the drive and extensive cooling of the rotor and/or the stator of the drive can be implemented. The at least one axial opening can be formed, for example, by a motor gap and/or the at least one axial groove can be formed by a channel-shaped free space between the stator and the housing. The at least one axial groove and/or axial opening can be formed in the housing and/or the drive unit.

[0026] According to a development, it has proven to be advantageous if, between the inlet and the first bearing unit, a flow path is branched into two flow paths connected in parallel, one of the flow paths extending through the hollow shaft portion and the other of the flow paths extending through the first bearing unit. The first bearing unit comprises a secondary bearing body, the secondary bearing body preferably being a roller bearing which is not sealed by seals and through which the other of the flow paths is preferably guided.

[0027] Furthermore, the inlet can be arranged in the machine housing directly upstream of the free end of the drive shaft, the inlet even more preferably being arranged approximately coaxially with the drive shaft in the longitudinal axis. The medium coming from the inlet is branched into two flow paths connected in parallel, the medium flowing through the first bearing unit in one path and the hollow shaft portion of the drive shaft in the other. As a result, in particular the secondary bearing body of the first bearing unit can be cooled and, in addition, the lubricant entrained with the medium can lubricate the secondary bearing body.

[0028] According to a second embodiment or a development of the present invention, the second bearing unit and the first spiral unit enclose a space, with at least one of the flow paths extending through the second bearing unit into the space.

[0029] The second bearing unit can comprise a main bearing body and a main bearing housing. At least one of the flow paths can be guided through the main bearing housing and/or through the main bearing body. In other words, the flow paths can be guided into the space through the second bearing unit exclusively through the main bearing housing or the main bearing body, or in parallel connection through the main bearing housing and the main bearing body.

[0030] However, at least one flow path is preferably guided through the main bearing body in order to cool the main bearing body and to lubricate the main bearing body by means of the lubricant entrained with the medium. According to a preferred embodiment, the main bearing body can be an unsealed roller bearing.

[0031] According to a preferred embodiment, the second bearing unit separates the machine housing into a drive portion and a suction region, the drive being arranged in the drive portion and the outer end regions of the first spiral unit and the second spiral unit being arranged in the suction region. The inlet opens into an inlet portion, the first bearing unit being arranged between the inlet portion and the drive portion. The medium is guided along the flow paths from the inlet portion through the drive portion and through the second bearing unit to the suction region or the outer end regions of the first spiral unit and the second spiral unit.

[0032] According to a development of the present invention, the drive shaft projects into the space through the second bearing unit, and a compensating mass securely connected to the drive shaft, and/or an eccentric drive for the first spiral unit, is/are arranged on the drive shaft in the space. An eccentric drive preferably comprises an eccentric portion which is designed as a driver on the drive shaft and which is coupled to the first spiral unit via an eccentric bearing body. Both the compensating mass and the eccentric drive can move in the space together with the drive shaft, as a result of which the medium present in the space is thoroughly stirred or “hurled around” and the lubricant entrained with the medium can be separated to a considerable extent.

[0033] According to a preferred embodiment, the second bearing unit can be cup-shaped or bell-shaped, the main bearing body and an axial bearing for the first spiral unit being provided on opposite sides of the second bearing unit in the longitudinal axis. The main bearing body is preferably arranged in the region of an apex of the bell-shaped second bearing unit and the axial bearing is arranged on the opposite end face.

[0034] According to a preferred development of the present invention, at least two flow paths connected in parallel lead from the inlet into the space, one flow path extending through the main bearing body and the other of the flow paths being guided through at least one entry opening formed as a through-hole in the main bearing housing. The two flow paths connected in parallel allow a pressure loss to be reduced while at the same time ensuring sufficient lubrication and/or cooling of the main bearing body.

[0035] The main bearing housing preferably has a plurality of entry openings, which are more preferably distributed symmetrically around the circumference. The entry openings are preferably arranged in a lateral surface of the main bearing housing, where, even more preferably, they are arranged in the longitudinal axis approximately centrally between the opposite sides of the second bearing unit.

[0036] According to a development of the present invention, the space has at least one exit opening and the exit opening defines a flow path which connects the space to the outer end regions of the first spiral unit and the second spiral unit or to the suction region.

[0037] Furthermore, it has proven to be advantageous for the at least one exit opening to comprise a first bore portion and a second bore portion, and for the first bore portion and the second bore portion to be arranged in an L or T shape. The first bore portion and the second bore portion are preferably formed along respective straight lines, which intersect at a common point of intersection. Even more preferably, with respect to the longitudinal axis the first bore portion is oriented in a radial direction and the second bore portion is oriented in an axial direction. The first bore portion is preferably designed as a through-bore and connects an outer lateral surface of the main bearing housing to the space. The second bore portion can be formed, for example, by a blind hole. Alternatively, at least one of the two bore portions can be formed in the main bearing housing by a primary shaping or reshaping process.

[0038] When the scroll machine is used as intended, an exit opening can preferably be arranged in such a way that lubricant can drain out of the space through the exit opening. For this purpose, in particular, an exit opening is arranged in an underside of the second bearing unit. Separated lubricant can accumulate in the region of the underside, where it can be entrained or drained by the flow through the exit opening.

[0039] In a further embodiment of the present invention, the at least one exit opening can be formed by an axial cut-out which extends from radially outside to radially inside and interrupts the axial bearing surface. The at least one axial cut-out is traversed when the first spiral unit moves along the orbital path, as a result of which the axial bearing can be continuously lubricated and cooled. The transition between the surface of the axial bearing and the at least one radially oriented axial cut-out can be provided with transition radii.

[0040] According to a development, an axial bearing element is arranged between the second bearing unit and the first spiral unit and is preferably arranged on the axial cut-outs which extend

from radially outside to radially inside and interrupt the axial bearing surface. The axial bearing element can preferably be an axial bearing plate.

[0041] Furthermore, it has proven to be advantageous for the at least one exit opening to be formed by a line in the second bearing unit, the line preferably comprising a first bore portion and a second bore portion, which are arranged in an L-shape. The first bore portion is preferably arranged in a radially oriented manner and the second bore portion is arranged in an axially oriented manner. The first bore portion and the second bore portion can preferably be incorporated into the second bearing unit or the main bearing housing, the first bore portion preferably penetrating the second bearing unit completely, while the second bore portion can be formed as a blind hole. The first bore portion consequently extends from the space to an outer lateral surface of the second bearing unit or the main bearing housing, where it is preferably closed by the machine housing or a machine housing portion. The second bore portion intersects the first bore portion and opens out in the outer end regions of the first spiral unit and the second spiral unit or in the suction region of the two spiral units.

[0042] According to a preferred development of the present invention, the at least one exit opening is, in relation to the longitudinal axis, circumferentially offset from the at least one entry opening in the space. Arranging the entry openings so as to be offset from the exit opening allows the medium to remain in the space for longer, as a result of which it is possible to increase the separation rate of the lubricant entrained in the medium in the space.

[0043] According to a preferred development of the present invention, the at least one exit opening, on the side of the second bearing unit facing the first spiral unit, is arranged partially or completely within a surface which is traversed by the first spiral unit when said unit moves completely along the orbital path. The at least one exit opening is thus partially or completely traversed at least once when the first spiral unit moves completely along the orbital path, as a result of which the axial bearing arranged between the second bearing unit and the first spiral unit can be lubricated.

[0044] In addition, it has proven to be advantageous for the first spiral unit to have, on the side facing the at least one exit opening, a recessed portion which is situated within a surface which traverses the at least one exit opening when the first spiral unit moves completely along the orbital path. The recessed portion prevents the first spiral unit from coming into direct contact with the at least one exit opening, which means that damage to the exit opening and/or to a surface of the axial bearing of the first spiral unit can be avoided.

[0045] According to a development of the present invention, at least one ring-pin coupling is provided, which prevents a complete rotation of the first spiral unit about the longitudinal axis. Such a ring-pin coupling comprises at least one ring-pin coupling pair, preferably a plurality of ring-pin coupling pairs, each having a pin that engages a corresponding abutment. In the annular abutment, the pin can perform a movement corresponding to the orbital path.

[0046] One of the flow paths is preferably guided through the ring-pin coupling, and even more preferably the abutment is formed in the second bearing unit and either perforates the second bearing unit completely or has a connecting bore which perforates the second bearing unit so that the flow path coming from the inlet can be guided through the abutment. The lubricant entrained by the refrigerant can collect in the ring-pin coupling for lubrication.

[0047] The at least one connecting bore can guide a flow path coming from the inlet through the second bearing unit to the suction region. The connecting bore does not necessarily have to be guided through a pair of ring-pin couplings or through the abutment thereof, as described above, but can also be guided beyond the pair of ring-pin couplings through the second bearing unit.

[0048] When the scroll machine is used as intended, the at least one pair of ring-pin couplings is arranged in such a way that lubricant can flow out of the drive portion through the ring-pin coupling toward the suction region. For this purpose, in particular, a ring-pin coupling is arranged in the region of an underside of the second bearing unit.

[0049] According to a development of the present invention, the second spiral unit is stationary.

The second spiral unit should therefore preferably not move relative to the first spiral unit and the machine housing when the scroll machine is being operated as intended.

[0050] According to a development or according to a further aspect of the present invention, a high-pressure chamber can be arranged in the housing. The inner end regions of the first spiral unit and the second spiral unit are connected to the high-pressure chamber via a passage, and the medium from the high-pressure chamber can leave the machine housing through the outlet.

[0051] The high-pressure chamber can be connected to an outlet via a pressure connection piece. The pressure connection piece can be arranged, in a plane transverse to the longitudinal axis, offset with respect to the passage and can more preferably be arranged in the pressure chamber along the longitudinal axis on the side opposite the passage. The offset arrangement between the passage and the pressure connection piece is intended to ensure that pressure pulsations are reduced and that the medium coming out of the passage cannot flow out of the scroll machine directly through the pressure connection piece.

[0052] In the high-pressure chamber a back-flow region can be provided which forces an S-shaped flow path from the passage to the outlet. The back-flow region facilitates dampening of pulsations and reduces pressure fluctuations in the medium discharged through the outlet.

[0053] According to a development, an intermediate bottom can be provided between the high-pressure chamber and the second spiral unit, the intermediate bottom together with the machine housing enclosing the high-pressure chamber. The intermediate bottom absorbs a large portion of the pressure load of the high-pressure chamber, as a result of which the second spiral unit is subjected to lower loads.

[0054] It has also proven to be advantageous for the back-flow region to be formed by a recess formed in the intermediate bottom on the side facing the high-pressure chamber and by the pressure connection piece, the pressure connection piece oriented toward the recess projecting into the high-pressure chamber.

[0055] Furthermore, it has proven to be advantageous for the pressure connection piece to be in active contact with the intermediate bottom in a contact region to form the flow region, and for the contact region, on an imaginary connecting line, to be arranged between the pressure connection piece and the passage in a plane perpendicular to the longitudinal axis.

[0056] According to a development, a non-return valve can be provided, which is arranged between the high-pressure chamber and the outlet. The non-return valve can be arranged both in the outlet and in the pressure connection piece, the non-return valve particularly preferably being inserted into the pressure connection piece in the form of a bushing. This results in a particularly compact and simple design.

[0057] A further aspect of the present invention relates to a refrigeration system comprising a scroll machine as described above.

Description

[0058] Two exemplary embodiments of the present invention are described in detail below with reference to the accompanying figures, in which:

[0059] FIG. 1 is a highly simplified and schematic representation of a refrigeration system comprising a scroll machine according to the invention;

[0060] FIG. 2 is an enlarged, simplified sectional view of a first exemplary embodiment of the scroll machine according to FIG. 1;

[0061] FIG. 3 is an enlarged detailed view of the scroll machine according to FIG. 2;

[0062] FIG. 4 is an enlarged, simplified sectional view of the scroll machine according to FIG. 1;

[0063] FIG. 5 is a sectional view of the scroll machine along the section line B-B in FIG. 4; and

[0064] FIG. 6 is a sectional view of the scroll machine along section line A-A in FIG. 2.

[0065] Identical or functionally identical parts or features are identified with the same reference signs in the following detailed description of the figures. In addition, not all identical or functionally identical components are provided with a reference number in the figures.

[0066] FIG. 1 shows a preferred embodiment of a refrigeration system 1 comprising a scroll machine 2. The refrigeration system 1 comprises the scroll machine 2 designed as a scroll compressor, a condenser 3, an expansion element 4, and an evaporator 5. A medium, preferably a refrigerant, flows through the refrigeration system 1 in the direction indicated by arrows, first from an outlet 12 of the scroll machine 2 then, in sequence, to the condenser 3, the expansion element 4, the evaporator 5, and finally back through an inlet 11 into the scroll machine 2.

[0067] A preferred embodiment of the scroll machine 2 shown in FIG. 1 is described below with reference to FIGS. 2 to 5.

[0068] FIG. 2 is a simplified sectional view of the scroll machine 2 according to FIG. 1. The scroll machine 2 has a machine housing 10, which is designated as a whole and oriented along a longitudinal axis X. The machine housing 10 can have a plurality of housing parts, the machine housing 10 having a first housing part 10' and a second housing part 10'' in the present exemplary embodiment.

[0069] From right to left, the inlet 11, a drive unit 400, a drive shaft 420, a first spiral unit 100, a second spiral unit 200, an intermediate bottom 50, a high-pressure chamber 30, and the outlet 12 are arranged in the machine housing 10 along the longitudinal axis X according to FIG. 2.

[0070] The first spiral unit 100 is coupled to the drive unit 400 via an eccentric drive 150 and the drive shaft 420.

[0071] The drive unit 400 preferably comprises an electric drive having a rotor 410 and a stator 415, the rotor 410 being rigidly coupled to the drive shaft 420.

[0072] The drive shaft 420 is oriented in the longitudinal axis X and the axis of rotation of the drive shaft 420 defines the longitudinal axis X in the exemplary embodiment shown. The drive shaft 420 has a first end portion and a second end portion on opposite sides in the longitudinal axis X.

[0073] In the first end portion, the drive shaft 420 is supported on the machine housing 10 by a first bearing unit 450 and in the second end portion by a second bearing unit 300. The rotor 410 of the drive unit 400 is arranged between the first bearing unit 450 and the second bearing unit 300.

[0074] The drive shaft 420 has a hollow shaft portion 424 which is oriented in the longitudinal X axis. The hollow shaft portion 424 can be designed as a blind hole and extends from a free end face of the drive shaft 420 in the first end portion toward the second end portion.

[0075] The drive shaft 420 also comprises a plurality of radial bores 428 which perforate the drive shaft 420 and connect the hollow shaft portion 424 to an outer lateral surface of the drive shaft 420.

[0076] According to FIG. 2, the drive shaft 420 can have at least one radial bore 428 between the first bearing unit 450 and the rotor 410 and/or between the second bearing unit 300 and the rotor 410. In the preferred embodiment shown, the drive shaft 420 has two radial bores 428 between the first bearing unit 450 and the rotor 410 and two radial bores 428 between the second bearing unit 300 and the rotor 410, which bores are arranged circumferentially symmetrically with respect to the drive shaft 420.

[0077] The first bearing unit 450 comprises a bearing holder 452 and a secondary bearing body 455. The bearing holder 452 can be formed by the machine housing 10. The secondary bearing body 455 can be designed as a roller bearing, which is preferably not sealed.

[0078] The first bearing unit 450 divides an interior of the machine housing 10 into an inlet portion and a drive portion, the inlet 11 opening into the inlet portion.

[0079] The inlet 11 is preferably arranged on an end face of the machine housing 10, with the inlet 11 even more preferably being arranged in alignment with the drive shaft 420—preferably directly—upstream of the free end face of the drive shaft 420.

[0080] The second bearing unit 300 comprises a main bearing housing 302 and a main bearing body 305. The main bearing body 305 can be designed as a preferably unsealed roller bearing. The

second bearing unit **300** further separates the interior of the machine housing **10** into the drive portion and a suction region **320**.

[0081] The second bearing unit **300** has a first side and a second side, the first side being nearest the drive unit **400** and the second side being nearest the first spiral unit **100**. The main bearing body **305** is arranged on the first side and an end face on the second side forms an axial bearing **190** for the first spiral unit **100**.

[0082] The second bearing unit **300** or the main bearing housing **302** can be bell-shaped or cup-shaped and enclose a space **380** together with the first spiral unit **100**.

[0083] The space **380** has a plurality of entry openings **370** which are arranged in the second bearing unit **300** or in the main bearing housing **302**. The entry openings **370** perforate the main bearing housing **302** and open into the space **380**. For example, the second bearing unit **300** can have four entry openings **370**, which, as shown in FIG. 5, can be arranged circumferentially symmetrically on a lateral surface, preferably approximately centrally between the first side and the second side.

[0084] Furthermore, the space **380** can have a plurality of exit openings **390** which are preferably situated in the second bearing unit **300** or the main bearing housing **302**. FIG. 3 shows an enlarged detailed view as shown in FIG. 2, from which it can be seen that the exit opening **390** can be formed from a first bore portion **392** and a second bore portion **394** in the main bearing housing **302**.

[0085] The first bore portion **392** is oriented substantially radially and extends from the space **380**. For manufacturing reasons, it can be advantageous for the first bore portion **392** to penetrate the main bearing housing **302** completely.

[0086] The second bore portion **394** is oriented substantially axially and connects the second side of the second bearing unit **300** or the main bearing housing **302** to the first bore portion **392**.

[0087] The second bearing unit **300** can have four exit openings **390**, which are preferably arranged circumferentially symmetrically. Furthermore, the exit openings **390** can be offset in the circumferential direction relative to the entry openings **370**—as indicated in FIG. 5.

[0088] With further reference to FIG. 2, it can be seen that the drive shaft **420** projects into the space **380** through the second bearing unit **300** or through the main bearing body **305**. A compensating mass **430** arranged on the drive shaft **420** is located in the space **380**.

[0089] Furthermore, it can be seen from FIG. 2 that the eccentric drive **150** is arranged in the space **380** and comprises an eccentric shaft portion **152** and an eccentric bearing body **155** arranged on the eccentric shaft portion **152**. The eccentric shaft portion **152** can be formed by the drive shaft **420**.

[0090] The first spiral unit **100** according to FIG. 2 has a first side **101** and has a second side **102** which is opposite the first side **101** in the longitudinal axis X. On the first side **101**, the first spiral unit **100** is supported on the second bearing unit **300** by means of the axial bearing **190**. The eccentric drive **150** is coupled to the first spiral unit **100** on the first side **101** and, on the second side **102**, a first spiral rib **110** is arranged which protrudes along the longitudinal axis X and forms a first spiral channel **120**.

[0091] A ring-pin coupling **350** is also provided, which comprises a plurality of ring-pin coupling pairs **351** (see FIG. 4 or 5). The ring-pin coupling pairs form the ring-pin coupling **350** which prevents full rotation of the first spiral unit **100** about the longitudinal axis X. The ring-pin coupling **350** couples the first spiral unit **100** to the second bearing unit **300** and comprises a pin **356** and comprises an abutment which can be formed by a recess **352** and a sleeve **354** situated in the recess **352**. In the abutment, the pin **356** can perform a movement corresponding to the orbital path.

[0092] According to the embodiment shown, the abutment can be formed in the second bearing unit **300** and the pins **356** protrude from the first side **101** of the first spiral unit **100** and engage the abutment of the second bearing unit **300**. A ring-pin coupling pair **351** can have a connecting bore

360 which perforates the second bearing unit **300** or the main bearing housing **302**.

[0093] It can also be seen from FIG. 2 that the first spiral rib **110** on the second side **102** of the first spiral unit **100** forms the spiral channel **120** with a spiral channel base. At the end face, the spiral rib **110** also has a first spiral rib tip, which can either have a seal or be designed as a flat tip. Furthermore, the first spiral channel **120** can have an inner end region **125** and/or an outer end region **126**.

[0094] The first spiral rib **110** is involute-shaped and extends from the inner end region **125** to the outer end region **126**. The inner end region **125** is located radially on the inside relative to the longitudinal axis X and the outer end region **126** is located radially on the outside relative to the longitudinal axis X. The at least one spiral channel **120** is U-shaped and is delimited in the radial directions by the spiral rib **110** or a spiral wall of the spiral rib **110** and by the spiral channel base.

[0095] The second spiral unit **200** can be stationary and has a first side **201** and has a second side **202** which is opposite the first side **201** in the longitudinal axis X. A second spiral rib **210** protrudes in the longitudinal axis X on the first side **201**, the second spiral rib **210** forming a second spiral channel **220**.

[0096] At the end face, the second spiral rib **210** also has a second spiral rib tip, which can either have a seal or be designed as a flat tip. Furthermore, the second spiral channel **220** can have an inner end region **225** and/or an outer end region **226**.

[0097] The second spiral rib **210** is adapted to the first spiral rib **110** and is also involute-shaped and extends from the inner end region **225** to an outer end region **226**. The inner end region **225** is located radially on the inside relative to the longitudinal axis X and the outer end region **226** is located radially on the outside relative to the longitudinal axis X. The at least one second spiral channel **220** is U-shaped and is delimited in the radial directions by the second spiral rib **210** or a spiral wall of the second spiral rib **210** and by the second spiral channel base.

[0098] As shown in FIG. 2, the first spiral rib **110** of the first spiral unit **100** and the second spiral rib **210** of the second spiral unit **200** engage or mesh with one another. The first spiral unit **100** can be moved through the drive unit **400** along an orbital path (not shown) relative to the second spiral unit **200**. A ring-pin coupling **350** prevents the first spiral unit **100** from rotating about the longitudinal axis X while moving along the orbital path.

[0099] Upon engaging or meshing with one another, the first spiral rib **110** engages the second spiral channel **220** and the second spiral rib **210** engages the first spiral channel **120**. The second spiral rib tip of the second spiral rib **210** sealingly interacts with the spiral channel base of the first spiral unit **100** and the first spiral rib tip of the first spiral rib **110** interacts with the spiral channel base of the second spiral unit **200**.

[0100] In a compressor, as the first spiral unit **100** moves along the orbital path between the first spiral unit **100** and the second spiral unit **200**, pressure chambers (not shown) are enclosed through which medium is transported from the outer end regions **126**, **226** to the inner end regions **125**, **225**. The outer end regions **126**, **226** together form the suction region **320**, from which the medium can be sucked into the spiral channels **120**, **220** in order to then be transported in closed pressure chambers (not shown) from the outer end region **126**, **226** to the inner end region **125**, **225**, the pressure chambers undergoing a continuous reduction in volume.

[0101] In an expander, as the first spiral unit **100** moves along the orbital path between the first spiral unit **100** and the second spiral unit **200**, pressure chambers (not shown) are enclosed which transport medium from the inner end regions **125**, **225** to the outer end regions **126**, **226**. During this process, the pressure chambers undergo a continuous increase in volume.

[0102] The medium is guided in the machine housing **10** along a plurality of flow paths from the inlet **11** to the outer end regions **126**, **226**, the medium being used along these flow paths to cool the components in the machine housing **10** and/or to lubricate them by means of entrained lubricant. In FIGS. 2-5, the flow paths are indicated by arrow lines; for better understanding only some of the arrow lines are marked with the reference sign "S".

[0103] The medium enters the machine housing **10** through the inlet **11** into the inlet portion. The medium then flows from the inlet portion into the drive portion.

[0104] In the inlet portion, the medium branches and flows along two flow paths connected in parallel toward the drive portion, one of the flow paths extending through the first bearing unit **550** and the other flow path being guided through the drive shaft **420** or through the hollow shaft portion **424** of the drive shaft **420**.

[0105] The medium can flow from the hollow shaft portion **424** into the drive portion through the radial bore **428** of the drive shaft **420**. The drive shaft **420** can have one or more radial bores **428** both between the first bearing unit **450** and the rotor **410** and between the rotor **410** and the second bearing unit **300**, as a result of which the flow path guided through the hollow shaft portion **424** branches several times and is used for cooling and/or lubrication, for example, of the rotor **410**, the first bearing unit **300** and/or the second bearing unit **450**.

[0106] Furthermore, it can be seen from FIG. 2 that the drive unit **400** has at least one axial opening **414** and/or at least one axial groove **418**, which each guide a flow path through the drive unit **400**. Each axial opening **414** and each axial groove **418** connects the two opposite end faces of the drive device **400**, as a result of which the drive unit **400** is “flushed” with the medium. Very effective cooling of the drive unit **400** can be realized by this measure.

[0107] The at least one axial opening **414** can be formed by a passing gap between the rotor **410** and the stator **415**.

[0108] From the drive portion, the medium then flows along a plurality of flow paths through the second bearing unit **300** into the suction region **320**.

[0109] The medium can be guided via the space **380** to the suction region **320** and can be guided via the connecting bore **360** in the ring-pin coupling **350** or in the at least one ring-pin coupling pair **351**.

[0110] The medium can be guided from the inlet **11** into the space **380** along two flow paths connected in parallel, one of the flow paths extending through the entry openings **370** and the other of the flow paths extending through the main bearing body **305**. In this way, the main bearing body **305** can be both cooled and lubricated by the lubricant entrained with the medium.

[0111] In the space **380**, the medium flows around the compensating mass **430** and the eccentric drive **150** of the first spiral unit **100**, as a result of which these components can be both cooled and lubricated.

[0112] The medium can leave the space **380**—as can be seen from the enlarged view according to FIGS. 3 and 5—through the exit openings **390**, which connect the space **380** to the suction region **320** or to the outer end regions **126**, **226** of the first spiral unit **100** and the second spiral unit **200**. The exit openings **390** open into the suction region **320** on the second side of the second bearing unit **300**, which side faces the first spiral unit **100**, each exit opening **390** preferably opening within a surface that is fully or partially traversed by the first spiral unit **100** when said unit moves completely along the orbital path. The positions at which the exit openings **390** open allow the first side of the first spiral unit **100** or the axial bearing **190** to be lubricated by the lubricant entrained by the medium.

[0113] In order to avoid damage to the exit opening **390** and/or the axial bearing surface of the first spiral unit **100**, the first spiral unit **100**—as shown in FIG. 3—can have, on the side facing the at least one exit opening **390**, a recessed portion **290** which is arranged within a surface which traverses the at least one exit opening **390** during a complete movement along the orbital path. In this way the exit opening **390** and the first spiral unit **100** do not come into contact.

[0114] A flow path can be guided from the drive portion and the suction region **320** through the ring-pin coupling **350** or the ring-pin coupling pairs **351**. Each ring-pin coupling pair **351** can have a connecting bore **360** (see FIG. 5) which guides a further flow path coming from the inlet **11** to the suction region **320**. For this purpose, the connecting bores **360** perforate the second bearing unit **300** and connect the first side of the second bearing unit **300** to the recess **352** in the first spiral unit

100. The pressure drop of the medium can be reduced by the connecting bores **360** in the ring-pin coupling pairs **351**.

[0115] According to a development (not shown), at least one connecting bore **360** can be provided which perforates the second bearing unit **300** between two ring-pin coupling pairs **351** and guides a flow path from the inlet **11** to the suction region **320**.

[0116] According to a development (not shown), the exit openings **390** can be formed by a plurality of radially oriented axial cut-outs arranged in the second side of the second bearing unit **300**.

[0117] The axial cut-outs can extend over the second side of the second bearing unit **300** and thus also over the axial bearing **190** formed between the second bearing unit **300** and the first spiral unit **100**. The lubricant entrained in the medium can contribute to lubricating the axial bearing **190** when the first spiral unit **100** traverses the recessed regions.

[0118] Furthermore, this development can comprise an axial bearing element which can be arranged in the form of a plate between the second bearing unit **300** and the first spiral unit **100**. The axial bearing element is preferably an axial bearing plate.

[0119] In the two exemplary embodiments described above, the high-pressure chamber **30** and the intermediate bottom **50** are arranged on the second side **202** of the second spiral unit **200**, the intermediate bottom **50** being arranged along the longitudinal axis X between the high-pressure chamber **30** and the second spiral unit **200**. The intermediate bottom **50** decouples the second spiral unit **200** from the pressure forces in the high-pressure chamber **30** and is supported with respect to the machine housing **10**.

[0120] The high-pressure chamber **30** is connected to the second spiral channel **220** via a passage **260**, the passage **260** comprising an outlet opening which is arranged in the region of the inner end regions **125**, **225**. The outlet opening, also known as the “discharge port”, is preferably formed in the inner end region **225** of the second spiral channel base and the passage **260** extends along the longitudinal axis X through a through-hole through the intermediate bottom **50** to the high-pressure chamber **30**.

[0121] The high-pressure chamber **30** is in turn connected to the outlet **12** and the medium can leave the scroll machine **2** through the outlet **12**.

[0122] The high-pressure chamber **30** is surrounded or housed by the machine housing **10** and the intermediate bottom **50** and has the outlet **12** through which the medium can leave the scroll machine. For this purpose, the machine housing **10** or the second housing part **10''** can be cup-shaped and have a recess, it being possible for the intermediate bottom **50** to close the high-pressure chamber **30** in the machine housing **10** or the second housing portion **10''** in the manner of a cover or plug. For this purpose, the shapes of the recess in the second housing portion **10''** and in the intermediate bottom **50** are adapted to one another, with both the recess and the intermediate bottom **50** preferably having a circular-cylindrical shape and being able to fit together precisely.

[0123] In order to avoid leakage between the intermediate bottom **50** and the machine housing **10**, sealing means can be provided there.

[0124] The intermediate bottom **50** has a first side and a second side, the first side being nearest the second spiral unit **200** and the second side being nearest the high-pressure chamber **30**. The intermediate bottom **50** comprises the through-hole through which the passage **260** extends.

[0125] In the embodiment shown, the intermediate bottom **50** has an annular projection protruding on the first side of the intermediate bottom **50** in the longitudinal axis X from the first side of the intermediate bottom **50** toward the second spiral unit **200**.

[0126] On the first side of the intermediate bottom **50**, there can be an axial securing means in the form of a securing ring which is fastened in the machine housing **10** and by means of which the position of the intermediate bottom **50** in the longitudinal axis X is determined. The axial securing means supports the intermediate bottom **50** on the side of the machine housing **10** facing the second spiral unit **200**, as a result of which the pressure forces from the high-pressure chamber **30** are substantially decoupled from the second spiral unit **200** and coupled into the machine housing

10.

[0127] The second spiral unit **200** can telescopically embrace the annular projection of the intermediate bottom **50** and for this purpose has, on the second side **202**, a first annular projection and a second annular projection, the first annular projection interacting with an inner lateral surface of the annular projection and the second annular projection interacting with an outer lateral surface of the annular projection of the intermediate bottom **50**.

[0128] The annular projection of the intermediate bottom **50** and the annular projections embracing the annular projection of the intermediate bottom **50**, on the second side **202** of the second spiral unit **200**, do not necessarily have to be provided, but rather represent a preferred development that can be used in particular when the scroll machine **1** has an injection system or an eco-port.

[0129] The annular projections of the intermediate bottom **50** and the annular projections of the second spiral unit **200** can form a radial bearing for the second spiral unit **200**.

[0130] As shown by the arrows in FIG. 2, the medium can reach the outlet **12** from the high-pressure chamber **30** via a pressure connection piece **40**, the pressure connection piece **40** preferably being arranged in such a way that the medium cannot flow directly from the passage **260** into the pressure connection piece **40**.

[0131] The pressure connection piece **40** projects from the side of the machine housing **10** facing the intermediate bottom **50**, in the direction of the intermediate bottom **50** and, as shown in FIG. 7, is offset from the passage **260** in a plane perpendicular to the longitudinal axis X.

[0132] In order to bring about a particularly effective reduction of pressure fluctuations in the high-pressure chamber **30**, a back-flow region **45** can be provided, which forces an S-shaped flow path S from the passage **260** through the pressure connection piece **40** to the outlet **12**, which is indicated in FIG. 2 by an arrow line.

[0133] The back-flow region **45** can have, on the second side of the intermediate bottom **50** facing the high-pressure chamber **30**, a preferably annular recess **59** (see FIG. 6) which together with the pressure connection piece defines the S-shaped flow path. For this purpose, the pressure connection piece **40** is in operative contact in a contact region **46** with the intermediate bottom **50** as shown in FIG. 6, the contact region **46**, on an imaginary connecting line, being located between the pressure connection piece **40** and the passage **260** in a plane perpendicular to the longitudinal axis X. As a result, the medium coming from the passage **260** must first be deflected in order to flow into the recess **59** and from the recess to reach the outlet **12** through the pressure connection piece **40**.

[0134] A non-return valve **48** shown in FIG. 2 can be arranged between the high-pressure chamber **30** and the outlet **12** and preferably comprises a bushing **49** that can be inserted into the pressure connection piece **40**.

LIST OF REFERENCE NUMERALS

[0135] **1** Refrigeration system [0136] **2** Scroll machine [0137] **3** Condenser [0138] **4** Expansion element [0139] **5** Evaporator [0140] **10** Machine housing [0141] **11** Inlet [0142] **12** Outlet [0143] **30** High-pressure chamber [0144] **40** Pressure connection piece [0145] **45** Back-flow region [0146] **46** Contact region [0147] **49** Bushing [0148] **50** Intermediate bottom [0149] **100** First spiral unit [0150] **101** First side [0151] **102** Second side [0152] **110** First spiral rib [0153] **120** First spiral channel [0154] **125** Inner end region [0155] **126** Outer end region [0156] **150** Eccentric drive [0157] **152** Eccentric shaft portion [0158] **155** Eccentric bearing body [0159] **190** Axial bearing [0160] **200** Second spiral unit [0161] **201** First side [0162] **202** Second side [0163] **210** Second spiral rib [0164] **225** Inner end region [0165] **226** Outer end region [0166] **220** Second spiral channel [0167] **260** Passage [0168] **290** Recessed portion [0169] **320** Suction region [0170] **300** Second bearing unit [0171] **302** Main bearing housing [0172] **305** Main bearing body [0173] **350** Ring-pin coupling [0174] **351** Ring-pin coupling pair [0175] **352** Recess [0176] **354** Sleeve [0177] **356** Bolts [0178] **360** Connecting bore [0179] **370** Entry opening [0180] **380** Space [0181] **390** Exit opening [0182] **392** First bore portion [0183] **394** Second bore portion [0184] **400** Drive unit [0185] **410** Rotor [0186] **414** Axial groove [0187] **415** Stator [0188] **418** Axial opening [0189] **420**

Drive shaft [0190] **424** Hollow shaft portion [0191] **428** Radial bore [0192] **430** Compensating mass [0193] **450** First bearing unit [0194] **452** Bearing holder [0195] **455** Secondary bearing body

Claims

1. A scroll machine (2) for a medium, comprising: a machine housing (10) having a longitudinal axis (X) and an inlet and an outlet for the medium, wherein, provided in the machine housing (10) along the longitudinal axis (X) is a drive unit (400) having a drive shaft (420) which is mounted on the machine housing (10) by a first bearing unit (450) and a second bearing unit (300); a first spiral unit (100) having a spiral channel (120) which is formed by a first spiral rib (110) and has an inner end region (125) and an outer end region (126), the first spiral unit (100) being provided in the machine housing (10); and a second spiral unit (200) having a spiral channel (220) which is formed by a second spiral rib (210) and has an inner end region (225) and an outer end region (226), wherein the first spiral unit (100) and the second spiral unit (200) engage one another to form pressure chambers, and wherein the second spiral unit (200) is provided in the machine housing (10), wherein the first spiral unit (100) is movable by means of the drive shaft (420) through the drive unit (400) along an orbital path relative to the second spiral unit (200), wherein an inlet (11) is in fluid communication with the outer end regions (126, 226) of the first and second spiral units (100, 200) and an outlet (12) is in fluid communication with the inner end regions (125, 225) of the first and second spiral units (100, 200), wherein the medium in the machine housing (10) flows along a plurality of flow paths from the inlet (11) to the outer end regions (126, 226) of the first and second spiral units (100, 200), and the drive shaft (420) has a hollow shaft portion (424) through which one of the flow paths extends, and, wherein a high-pressure chamber (30) is arranged in the machine housing, in that the inner end regions (125, 225) are connected to the high-pressure chamber (30) via a passage (260) and are connected to the outlet (12) via the high-pressure chamber (30), and in that in the high-pressure chamber (30) a back-flow region (45) is provided which forces an S-shaped flow path in the high-pressure chamber (30).
 2. The scroll machine (2) according to claim 1, characterized in that an intermediate bottom (50) is provided between the high-pressure chamber (30) and the second spiral unit (200) and in that the back-flow region (45) is formed by a recess (59) formed in the intermediate bottom (50) on the side facing the high-pressure chamber (30) and by a pressure connection piece (40) projecting toward the recess (59).
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